Geography of X* Energy, Vitality, and Happiness Across US Counties

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We use X (Twitter) data to study Energy, vitality, and happiness Across US Counties. Energy, vitality, and happiness is measured based on human expression on X. We use 9 metrics: 3 measuring energy/vitality: positive functioning, vitality, and having energy; and 6 measuring happiness: life satisfaction, happiness, peace with thoughts and feelings, expressing optimism, purpose in life, having hope. Cronbach's Alhpa scores of 0.85-0.92 indcate high internal consistency and reliability of our novel scales. One contribution is the first study of geography of energy/vitality, where we find a very clear urban-rural gradient with energy/vitality raising from its lowest levels for large counties to highest level in small counties. Energy/vitality may be actually more important than happiness and at least as important, and it is striking that we know nothing about its geography. There is literature, on the other hand, on geography of happiness–as found earlier in the literature cities tend to be less satisfied with life, but overall on an index and other happiness measures, there is an important qualification. It is not the very smalles places in 1st population decile that are the happiest (they are actually the least happy along with the largest counties in 10th decile), but small counties in deciles 2-4 (5.1-18.5k).

TODO sdd key interpretations of regions and largest cities. Caveats and limitations remain.

While X wellbeing expressions tend correlate moderately at about .6 with representative survey data, geotagged tweets are overrepresented among certain populations.

1 Introduction

Terms SWB, happiness, and life satisfaction are usually used interchangeably in happiness and social indicators literatures. But specifically they mean rather cognitive global subjective self-evaluation of one's life as a whole rather than momentary affective happiness. Here it rather is the latter, momentary affective happiness, as human expression is recrded via tweets on X.[1]

Much ink has been spilled on geography of Subjective Wellbeing (SWB), including in the US, in particular there are studies across counties (e.g., Okulicz-Kozaryn and Mazelis 2018, Winters et al. 2015, Okulicz-Kozaryn et al. 2024), and across urban rural continuum (Okulicz-Kozaryn 2024, 2023, Okulicz-Kozaryn and Valente 2020). Major findings have been that humans are less satisfied with their lives in populated dense North East, but also in poorer Midwest, and some of the lowest life satisfaction scores in addition to largest cities such as NYC and Philadelhia, have been reorded in South, in Particular in poor Alabama, Mississipi, and Louisiana. Greatest life satisfaction, on the other hand, has been found in sparsely populated North and West, in states such at Nebraska and Utah, but not so much in overpopulated California.

That has been in SWB and geographic literatures. One limitation of these literatures is sole use of survey data such as GSS and BRFSS and typically just one survey item about life satisfaction such as "How satisfied are you with your life as a whole?" But human wellbeing can be measured in many other ways. Here we use across about 3k US counties 9 metrics: 6 measuring happiness, and 3 measuring energy/vitality. This is the main contribution of our study to expand on measurement of county and urban-rural wellbeing. In the process of doing so we also contribute by connecting happiness and geography literatures (summarized in Okulicz-Kozaryn 2023, Okulicz-Kozaryn and Valente 2020)[2] with X literature, which seems to be mostly in computer science.

*Twitter

[1]Reliability and validity of twitter data discussion is postponed to section "Data."

[2]For elaboration see Ballas and Rijnks (2023), Ziogas et al. (2023), Weckroth et al. (2022), Ziogas et al. (2020), Ballas (2013), Knies (2026), Sørensen (2024, 2014), Lenzi and Perucca (2022, 2020, 2016c,a,b).

2 X geography of wellbeing literature

There is a quite separate X (Twitter) literature, that started around 2010 shortly after Twitter launched in 2006. One early example on overall geography of X is widely cited Leetaru et al. (2013), but it does not study happiness nor energy/vitality nor any other wellbeing metrics. Various subsecquent studies did measure boradly defined wellbeing including energy/vitality and happiness, but they differ in important ways from our study except pioneering Mitchell et al. (2013) that we discuss towards the end of this section.

There is an intersting and related doctoral dissertation (Gupta 2018), but analyzes wellbeing geographies in UK.

Chai et al. (2023) produced "twitter Sentiment Geographical Index (tSGI), a location-specific expressed sentiment database with SWB implications", but it doesnt really analyze energy/vitality nor happiness and has a global, not the US, focus.

There are neighborhood level studies in NYC studying happiness and poverty (Tan and Guan 2021) and proximity to parks (Plunz et al. 2019). And one in Detroit MI analyzing happiness and neighborhood conditions (Park et al. 2021). And another analyzing happy tweets, healthy food references, and physical activity across neigborhoods in Salt Lake, San Francisco and New York (Nguyen et al. 2016). Anothother study analyzes sentiment across spatiotempoprtal dimensions (land use and times of the day) across Massachusetts (Cao et al. 2018).

Then there are several studies that are closer to ours. Hollander and Renski (2022) studies cities, like us, but excludes rural areas unlike us, and only analyzes 50 declining cities matched with 50 growing/stable cities and only using 300k tweets.

Even though Hu et al. (2019) may seem similar to our study as it bradly asses sentiment and positive emotions across us counties, it is quite different. Their approach and framing are very different. Rather than focusing on geography of wellbeing like us, they explore tweeter topics and correlations in space. Rather than energy/vitality or happiness, they study variuos topics such as birthdays, parties, and great time on sunday and focus on spatial correlations rather than levels like us.

The closest to what we do is Mitchell et al. (2013) like us focus on geography within the US, but they study states (we study counties), and they study also some cities like us, but only a limited set, while our data covers virtually all of united states including rural counties. They only have 80m words generated in 2011 on X, but we use a much larger dataset of over 1b tweets 2010-2019. While they also show some results for all of the US like us in their map in figure 5, it is difficult to read for less populated areas. Another limitation that we improve on is the use of multiple energy/vitality and happiness metrics as opposed to just one happiness metric in their study. Also note the Atlantic article that discuss the arxiv version of their study.[3] We discuss Mitchell et al. (2013) more as we present our results. Again, note that Mitchell et al. (2013) is one of the pioneering studies done over 10 years ago.

Last, but not least there is an arxiv preprint (Iacus and Porro 2025) by producers of the data that we use. We both study US geography of happiness, and we both use life satisfaction and happiness metrics. But here similarities end. In addition to life satisfaction and happiness metrics we use 7 other measures. Iacus and Porro (2025) measure rural-urban with USDA ERS 9 Rural-Urban Continuum Codes; we use population deciles. Iacus and Porro (2025) doesn't focus on any US counties, except 23 most rural counties in their table 4; we focus on largest 20 counties instead. Finally, Iacus and Porro (2025) in addition to geography also focus on politics and income, while we focus just on geography and dig deeper. Thus, despite using 2 variables from the very same dataset, our studies differ in critical ways.

Curiously Kjell et al. (2023) shows a map simialr to our county estimates in fig 5.1C and cites Jaidka et al. (2020). Jaidka et al. (2020) focuses on comparison of twitter happiness with Gallup data but doesn't analyze by urban-rual–and county-level estimates cited in Kjell et al. (2023) are not there, not even in supplemtary material, hence it is difficult to discuss. But there are differences: we have multiple measures and do analysis by urban rural. And, crucially, the results that overlap, county level happiness maps–the findings differ. We discuss more when discussing our maps. A useful contribution of Jaidka et al. (2020) is a finding that wellbeing X measures correlate well at about .6 with survey data.

Another stduy by Jaidka (Jaidka et al. 2021), like us, analyzes of US counties and urban-rural, but in terms of stress.

[3] https://www.theatlantic.com/technology/archive/2013/02/the-geography-of-happiness-according-to-10-million-tweets/273286/

Overall, while there is X geography of happiness literature, it either focus on different geographies (other countries, or other aggregations such as state or neghborhoods) and different (e.g., stress) or more limited metrics such as one measure of happiness only.

3 Data

Data are from `https://dataverse.harvard.edu/file.xhtml?fileId=13094135&version=3.1` using file `flourishingCountyYear.csv` Description of the data is at `https://arxiv.org/pdf/2511.03915` (Iacus et al. 2025):

> Here we introduce the Human Flourishing Geographic Index (HFGI), derived from analyzing approximately 2.6 billion geolocated U.S. tweets (2013-2023) using fine-tuned large language models to classify expressions across 48 indicators aligned with Harvard's Global Flourishing Study[...].(Iacus et al. 2025, p. 1)

Note that while data description says 2013-2023, in the actual dataset we also found observations 2010-2012, but it was only 7.9% of the whole dataset, out of which miniscule <1% before 2012.

We cut out 2020-2023 due to Covid19 pandemic, but that still leaves us with over 1b tweets. Pandemic affected US geographies very differently and thus would be likely to bias our results–low wellbeing would have been attributed to place, while it was due to Covid19.

We also dropped county-year-variable cells with <5 valid tweets. If there were very few tweets in a cell, the county-year-variable average could be often more extreme than average from larger and more representative sample and thus bias disproportionatelly the results.

One could frown upon Twitter data, but here there are actually reasons to be optimistic. There are > 10 wellbeing measures, and data covers 2012-2019, having it collapsed for each county over 8 years makes sample size bigger and provides more reliability.

The reliability/validity seems acceptable, again wellbeing X measures correlate at about .6 with survey data (Jaidka et al. 2020). But reliability/validity it probably not high. While some are excited and overly positive (e.g., Karamouzas et al. 2022), others raise important points specifically regarding geotagged tweets.

Geotag users[4] are not representative of the US population. Number of geotag users is predicted by younger users, users of higher income, and users in urbanized areas, Hispanic/Latino users and Black users, and east or west coast (Malik et al. 2015).

And maybe the most critical Jensen (2017), and especially relevant here as it is a critique of our main reference: Mitchell et al. (2013): TODO elaborate in detail!!!

We dont get here into twitter data processing and technicalities of natural language processing–it is not our area and we didnt do this–again, we just downloaded pre-processed data and the documentation is in Iacus et al. (2025). We want to focus here on the meaning of the variables instead.

It is important to note a caveat: in the choice of variables we just used our judgement, there is no literature or agreed upon set of metrics here. We are encouraged by high Cronbach's Alhpa scores that indcate high internal consistency and reliability or our novel scales.

we made indices by simply averaging acress indicartors –these vatrables go veywell together at high cronbachs alpha: we have 9 indicators (3 energy/vitality and 6 happiness). 3-item energy/vitality index `eneI` has a high (good) Cronbach's alpha of 0.85. 6-item happiness index `swbI` has a very high (excellent) Cronbach's alpha of 0.92.

$$'eneI' = mean('posfunct',' vitality',' energy') \tag{1}$$

[4]For geographic analysis such as ours across counties and urban-rual spectrum one needs geographic location, i.e., geotagged tweet.

$$'swbI' = mean('lifesat', 'happiness', 'innerpeace', 'optimism', 'purpose', 'hope') \tag{2}$$

- Vitality. Feelings of strength and activity (`vitality`).
  low: "Still recovering from a cold, running on empty."
  med: "Tired but determined to finish this week strong."
  high: "The energy today is unreal–everything feels possible!"

- Having energy. Reports of feeling (or not) energetic (`energy`).
  low: "Can't seem to get moving this morning."
  med: "A little sluggish but getting things done."
  high: "So full of energy I could run another mile!"

- Positive functioning. Feeling capable and effective (`posfunct`).
  low: "Everything feels like too much lately."
  med: "Getting back into a productive rhythm again."
  high: "Handled a dozen tasks today–completely in the zone!"

- Life Satisfaction. Satisfaction with one's life as a whole (`lifesat`).
  low: "My living situation is exhausting and unfair."
  med: "Things are slowly falling into place–can't complain too much."
  high: "Watching the sunset from my porch tonight, feeling truly content."

- Happiness. The text expresses some level of happiness (`happiness`).
  low: "Mondays always feel endless and gray."
  med: "Trying to notice at least one small thing that makes me smile today."
  high: "We reached our team goal today–feeling genuinely proud and joyful!"

- Expressing optimism. Optimism about one's condition in the medium run (`optimism`).
  low: "I don't expect much to improve soon."
  med: "Maybe next month will bring better news."
  high: "Things are finally turning around–I can feel it!"

- Peace with thoughts and feelings. General feeling of calm acceptance (`innerpeace`).
  low: "Can't quiet my thoughts enough to rest."
  med: "Learning to take things as they come."
  high: "Feeling calm and centered after a long day."

- Purpose in life. Acting with a sense of purpose (`purpose`).
  low: "I'm not sure what I'm doing any of this for."
  med: "Trying to find direction and meaning in small steps."
  high: "Every action today felt aligned with what I believe in."

- Having hope. Hope about the future (`hope`).
  low: "Hard to see a way forward."
  med: "Things might improve if we stay patient."
  high: "Tomorrow will be brighter–I truly believe it."

it is har to believe for us urban america, that there exists rurla america (Hanson 2015), and that in fact there are about 300 counties (yes whole counties not just villages) with pop < 5k!

4 Results

4.1 Urban-rural gradient

We start with urban-rural happiness gradient (Berry and Okulicz-Kozaryn 2011) in table 1 looking at energy/vitality. There is a remarkable consistency: the urban rural happiness gradient is raising from its lowest lewels in largest counties (color coded with red) to highest levels in smallest counties (green). The only one exception is in `energy` column in cell at population decile of "25.6k-36.7k." An important pattern is that it is not only the very largest counties in the top 10th decile in last row that have lowest energy/vitality, but also other large counties in 7th-9th deciles (36.7k-204.9k). So lowest energy/vitality is not only in the very largest counties in 10th decile, but already in 7th decile adnn higher (>36k. Likewise highest energy/vitality is not only in the very smallest counties in 1st decile (<5.1k), but already in 3rd decile and below (<13.5k), and especially in 2nd decile and below (<8.9k).

One nuance is that `posfunct` and `vitality` columns show the lowest levels (dark red) not in the very largest counties in the last row "204.9k-10077.3k" but in slightly smaller but still above the median population size rows.

Also note quite large substantive differences. Energy/vitality index `eneI` is 12% (.35/.31) difference between 1st and 10th deciles. `Postfunct` and `vitality` are even bigger percentage differences.

population 2015 decile	eneI	posfunct	vitality	energy
88-5.1k	0.35	0.37	0.26	0.43
5.1k-8.9k	0.35	0.35	0.26	0.43
8.9k-13.5k	0.34	0.33	0.25	0.43
13.5k-18.5k	0.33	0.32	0.24	0.43
18.5k-25.6k	0.32	0.31	0.22	0.42
25.6k-36.7k	0.32	0.31	0.21	0.43
36.7k-52.9k	0.31	0.30	0.20	0.42
52.9k-92.6k	0.31	0.30	0.20	0.42
92.6k-204.9k	0.31	0.29	0.20	0.42
204.9k-10077.3k	0.31	0.30	0.21	0.42

Figure 1: Note we stick to `lifesat` tag (life satisfaction) as in the origan dataset, but again it is rather momentary affective happiness than global cognitive evaluationn of one's life as a whole, life satisfaction term used in SWB literature.

Table 2 repeats the exercise but for happiness index `swbI` and individual measures. Here as expected and shown in the literature (Berry and Okulicz-Kozaryn 2011) in second colored column `lifesat`, life satisfaction raises from its lowest level for largest counties in last decile of population to its highest level for the smallest counties in first decile of population. But this pattern is only observed for this one column. Overall, in all other column the pattern tends to be of lowest wellbeing in 10th decile as expected, but also in the 1st decile. Still, the urban-rural happiness gradient hypothesis (Berry and Okulicz-Kozaryn 2011) holds as theh counties with highest wellbeing scores are the smallest counties in 2nd, 3rd, and 4th population size deciles (5.1k-18.5k), and some of the lowest wellbeing scores are also in the second highest 9th population decile. Overall, we find lowest happiness in 1st and 10th population deciles and highest in 2nd-4th population deciles. While as expected smaller 2nd-4th deciles counties tends to be happier and largest 9th-10th deciles counties tend to be less happy, it is also the very smallest counties in 1st deciles that are unhappy.

`Lifests` is as expected in the literature, and do note huuge urban-rual happiness gradient change all the way from .06 for 10th decile to .15 for the 1st decile, >2x difference. But `happiness` is almost the other way round from lowest in smaller counties to larger in larger counties (but not very largest), slightly an inverted U-shape with maximum at 6th decile (25.6k-36.7k) and `optimism` is very similar. Hope has a clear inverted U-shape with maximum at 6th decile and minimums at 1st and 10th deciles. Still, overall from the happiness index swbI (and innerpeace) it is not the very smallest ones like <5.1k, but also small ones already at <18.5k.

There are also some nuances, for instance 6th decile (25.6k-36.7k) scores highest on happiness, optimism, and hope[5]. Hence, for these measures there is evidence of inverted U-shape: highest happiness is around median population size and lowest in smallest and largest counties.

population 2015 decile	swbI	lifesat	happiness	innerpeace	optimism	purpose	hope
88-5.1k	0.24	0.15	0.24	0.13	0.15	0.38	0.40
5.1k-8.9k	0.27	0.14	0.27	0.16	0.19	0.41	0.43
8.9k-13.5k	0.27	0.12	0.29	0.16	0.21	0.42	0.44
13.5k-18.5k	0.27	0.11	0.29	0.16	0.21	0.40	0.43
18.5k-25.6k	0.26	0.10	0.29	0.15	0.21	0.39	0.43
25.6k-36.7k	0.26	0.09	0.30	0.15	0.22	0.39	0.44
36.7k-52.9k	0.25	0.08	0.29	0.14	0.20	0.37	0.42
52.9k-92.6k	0.25	0.08	0.29	0.14	0.20	0.37	0.43
92.6k-204.9k	0.24	0.07	0.28	0.13	0.20	0.36	0.42
204.9k-10077.3k	0.24	0.06	0.28	0.13	0.19	0.35	0.40

Figure 2: This figure was pulled directly from a web URL.

4.2 Across counties

We turn to mapping energy/vitality index eneI and happinessindex swbI[6] across counties.

eneI is in figure 3.

South does not surprise. As expected, the energy/vitality is low in South: epseically in Missisipi and Loiusiana, but also in most parts of Texas and Alabama. Second comes Midwest, no surprise there either. Ohio is the lowest almost all low energy/vitality, and then most of Pennsylvania and and about half of West Wirginia and Michigan–in South Michigan is low, North Michigan is high.

Also about half of Oklahoma and Kansas are low on energy/vitality. On West coast, perhaps surprisingly about half or Oregon is low on energy/vitalty. And in Georgia and Arkansas about third.

As opposed to dull South and Midwest, there is definitely more energy/vitality in North East, for instance in New Jersey–one would think. But our results disagree–there is not much difference, if any. Dull and low energy South does contrast with upbeat and high energy North East–it is just that the energy may not be that good after all–stressed, hectic and neurotic. This concludes our general list of low energy/vitality states, but do note many red spots theought the US and in other states.

Turning now to high energy/vitality: in general west and north but not coastal/pacific west: not OR and CA Colorado seems the highest energy/vitality, but also Utah, Wyoming, Montana, and Idaho.

Note the maps with detailed picture more color bins are in appendix. Overall they are very similar and simplified exposition is presented here but the more important differences are: the extremes remain the same: dark red and dark green but there is much more gradation for mid-levels from light green through yellow to light red.

The happiness map in figure 4 broadly agrees with energy/vitality map in figure 3, and the correlation between the two is .83, for a scatterplot see appendix. There are few differences, for instance Tennesee, Arkansas, and Missouri have more happiness than energy/vitality, North East, too, except New Jersey that is equally unhappy as it has low energy/vitality. As opposed to above energy/vitality index in the above map 3, the below happiness map in figure 4 is not enirely new.

Results in figure 4 broadly agree at state level with Mitchell et al. (2013). Much of the results in our figure 4 also seem to agree

[5]Along with 3rd decile.

[6]Here as maps show, it is only contigious US, but in other analyses there were few additional observations from Alaska and Hawaii.

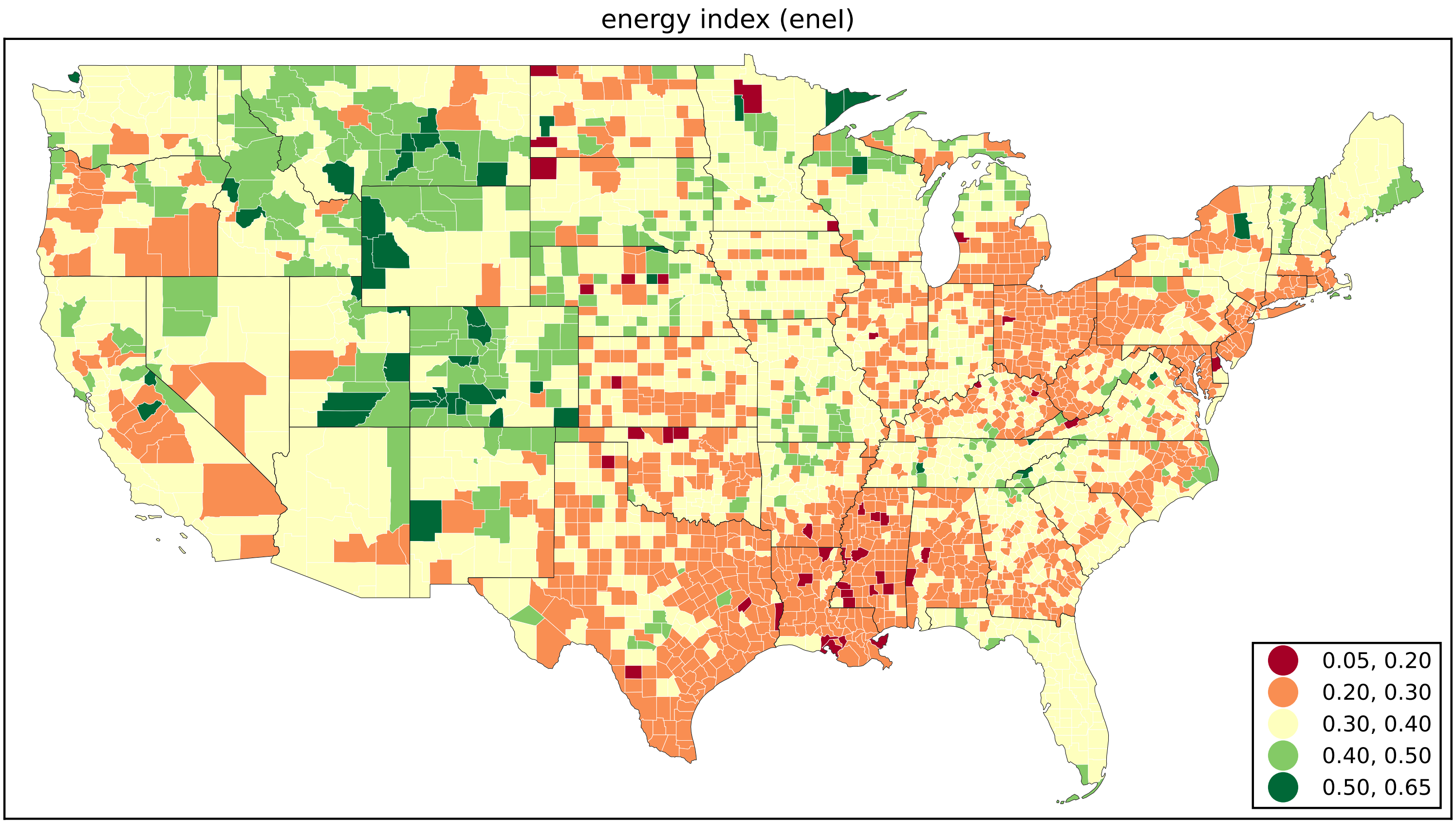

Figure 3: Energy/vitality index.

at county level with Mitchell et al. (2013), but it is difficult to say as map averages at county level, while Mitchell et al. (2013) map closer to actual tweet locations and there are multiple points per county and it is difficult to estimate the averge.

Let's also discuss an Atlantic article that covers the arxiv version of their study.[7].

¿¿¿

Kjell et al. (2023) shows a map simialr to our county estimates in figure 5.1C citing Jaidka et al. (2020). It looks like their sample of 1.5b is close to ours. But the results are different possibly due to their "interpolation of missing counties provided through a Gaussian process model using demographic and socioeconomic similarity between counties" (Kjell et al. 2023, fig. 5.1C caption). Alabama, Louisiana, and Missisipi are not unhappy, neither is North East, but Midwest is least happy (we find it unhappy but not as unhappy as Alabama, Louisiana, and Missisipi.

But maybe the most striking difference is that urban counties marked in their figure 5.1C with city names all appear to be the very happiest counties of all (dark green). Except Los Angeles and Miami (2nd shade of green, still very happy), Phoenix (yellow; avergage happiness), and per New York and New Orelans cannot really tell as the black city dots cover polygons. Curiously, World Happiness Report and Gallup argue urban happiness despite data showing the contrary–see discussion in Okulicz-Kozaryn and Valente (2021). For instance see `https://news.gallup.com/opinion/gallup/315857/degree-urbanisation-effect-happiness.aspx`.

4.3 Zooming in on largest cities

again figure 5.1C with city names all appear to be the very happiest counties of all (dark green).

while the black dot covers most or all of the city-counties, but even the metropolitan area bordering counties that are visible and

[7]https://www.theatlantic.com/technology/archive/2013/02/the-geography-of-happiness-according-to-10-million-tweets/273286/

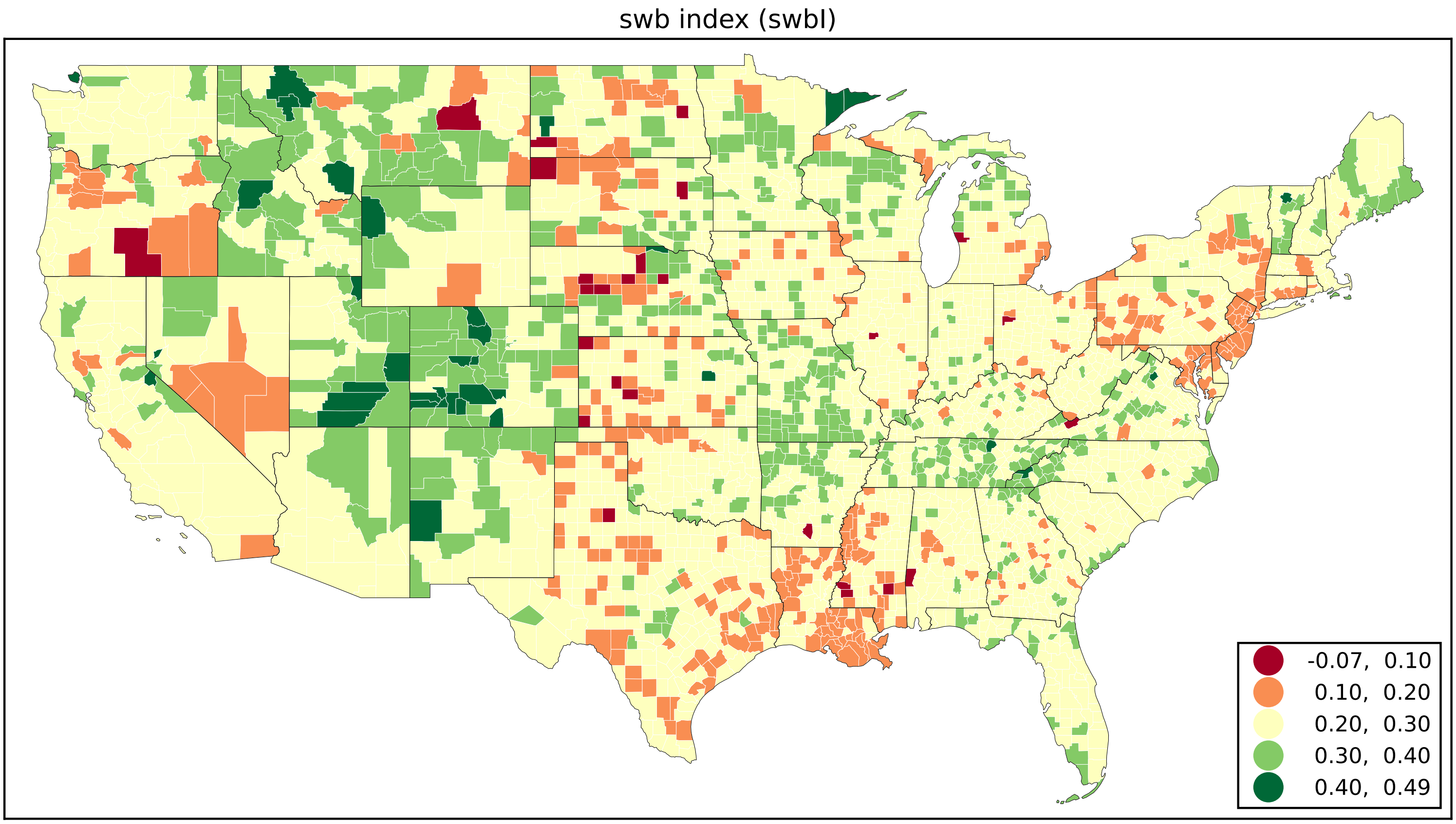

Figure 4: This figure was pulled directly from a web URL.

very happy, which already is surprising

Except Los Angeles and Miami (2nd shade of green, still very happy), and Phoenix (yellow; avergage happiness), and per New York and New Orelans cannot really tell as the black city dots cover polygons. Likewise difficult to tell per other cities due to black big dots, but it seems they are from West to East: Seattle, Los Angeles, Salt Lake City, Pheonix, Denver, Minneapolis, Chicago, Houston, Atlanta, Washington, and Miami

It is important to keep in mind that all data including population pertains to a county, not city, for instance Santa Ana is only about 300k but Orange county is about 3m. Still using cities for ease of iterpretations.

First,for ease of interpretation summarizing all indicators as happy or unhappy as for most counties there is a clear pattern.

First 4 numeric columns are energy/vitality (enel, posfunct, vitality, energy) and subsequent 6 numeric columns are happiness (swbI lifesat happiness innerpeace optimism purpose).

Happy counties are sunny places, but outside of unhappy Texas, and many coastal like Los Angeles, Phoenix, San Diego, Santa Ana, Miami, Las Vegas, San Jose and Ft Lauderdale. Also happy are cold but coastal Seattle and Manhattan (New York County).

Detroit and Houston are the least happy.In addition to Houston, there are 2 otherunhappy Texas cities, Dallas and Forth Worth. NYC Queens and Brooklyn ar enot too happy either in sharp contrast to very happy Manhattan.

Then few interesting nuances. For instance 3 Texas cities Dallas, Forth Worth (neighbors), and San Antonio score poorly on energy/vitality (first 4 numeric columns) but quite well on happiness (subsequent 6 numeric columns).

At the bottom comparison to deciles from earlier figures (note that comparison colors in the table differ as they are scaled to current table). decile 1 tends to be higher than mean of deciles 2-4

of the largest metropolises some score better or worse than decile averages

Highest and lowest bot energy/vitality and happiness tends to be in small counties, below median population size (see appendix).

city	STNAME	CTYNAME	pop.(m)	eneI	posfunct	vitality	energy	swbI	lifesat	happiness	innerp
Los Angeles	California	Los Angeles County	10.1	0.34	0.34	0.26	0.44	0.26	0.12	0.29	
Chicago	Illinois	Cook County	5.2	0.32	0.30	0.21	0.46	0.26	0.11	0.30	
Houston	Texas	Harris County	4.6	0.28	0.26	0.17	0.40	0.17	0.03	0.16	
Phoenix	Arizona	Maricopa County	4.2	0.34	0.33	0.25	0.45	0.27	0.11	0.31	
San Diego	California	San Diego County	3.3	0.36	0.35	0.29	0.45	0.28	0.14	0.34	
Santa Ana	California	Orange County	3.1	0.37	0.36	0.29	0.46	0.29	0.15	0.35	
Miami	Florida	Miami-Dade County	2.7	0.36	0.36	0.28	0.45	0.28	0.14	0.32	
Brooklyn	New York	Kings County	2.6	0.32	0.31	0.23	0.44	0.21	0.06	0.28	
Dallas	Texas	Dallas County	2.6	0.28	0.27	0.18	0.40	0.22	0.04	0.26	
Riverside	California	Riverside County	2.3	0.31	0.30	0.21	0.41	0.24	0.10	0.29	
Queens	New York	Queens County	2.3	0.30	0.28	0.19	0.42	0.18	0.01	0.24	
Seattle	Washington	King County	2.1	0.36	0.35	0.28	0.46	0.28	0.11	0.33	
San Bernardino	California	San Bernardino County	2.1	0.30	0.29	0.20	0.40	0.23	0.08	0.28	
Las Vegas	Nevada	Clark County	2.1	0.37	0.35	0.28	0.47	0.28	0.15	0.34	
Fort Worth	Texas	Tarrant County	2.0	0.27	0.25	0.17	0.39	0.21	0.04	0.25	
San Jose	California	Santa Clara County	1.9	0.35	0.34	0.26	0.46	0.28	0.12	0.33	
San Antonio	Texas	Bexar County	1.9	0.28	0.27	0.17	0.40	0.22	0.05	0.27	
Fort Lauderdale	Florida	Broward County	1.9	0.35	0.34	0.27	0.43	0.26	0.12	0.30	
Detroit	Michigan	Wayne County	1.8	0.25	0.24	0.14	0.38	0.19	-0.01	0.24	
Manhattan	New York	New York County	1.6	0.38	0.37	0.28	0.48	0.28	0.14	0.35	
		decile 1 (<5.1k)		0.35	0.37	0.26	0.43	0.24	0.15	0.24	
		mean of deciles 2-4 (5.1-18.5k)		0.33	0.32	0.23	0.43	0.26	0.10	0.29	
		decile 10 (>204.9k-10077.3k)		0.31	0.30	0.21	0.42	0.24	0.06	0.28	

Figure 5: Energy/vitality and happiness for 20 largest counties. Counties are sorted on population. At the bottom for comparison averages for 1st decile, 2-3rd deciles, and 10th decile.

5 Conclusion and discussion

This is the first study of geography of energy/vitality, where we find a very clear urban-rural gradient with energy/vitality raising from its lowest levels for large counties to highest level in small counties. Energy/vitality may be actually more important than happiness and at least as important, and it is striking that we know nothing about its geography.

ONLINE APPENDIX

[note: this section will NOT be a part of the final version of the manuscript, but will be available online instead]

5.1 Additional results

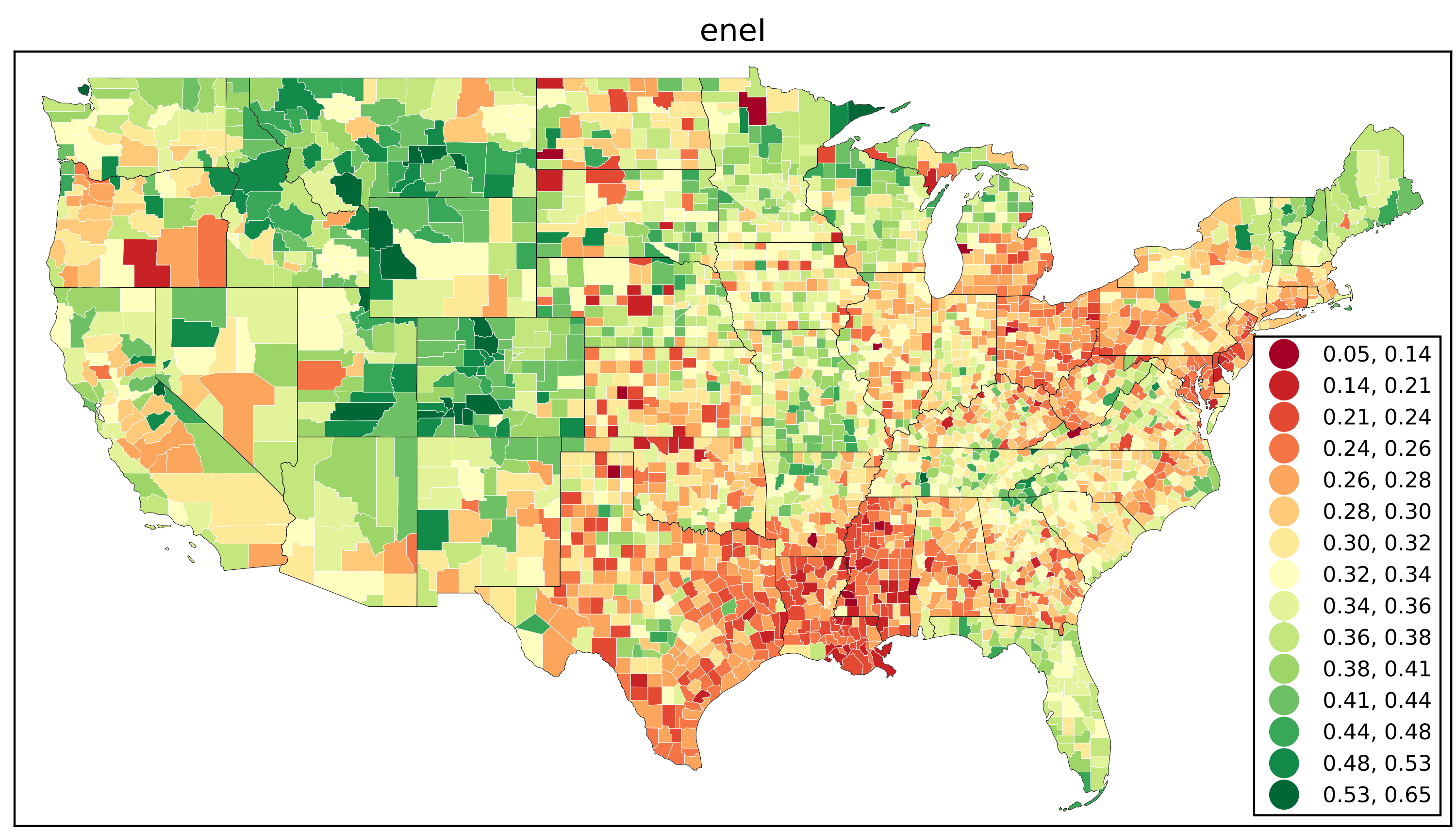

Figure 6: more classes This figure was pulled directly from a web URL.

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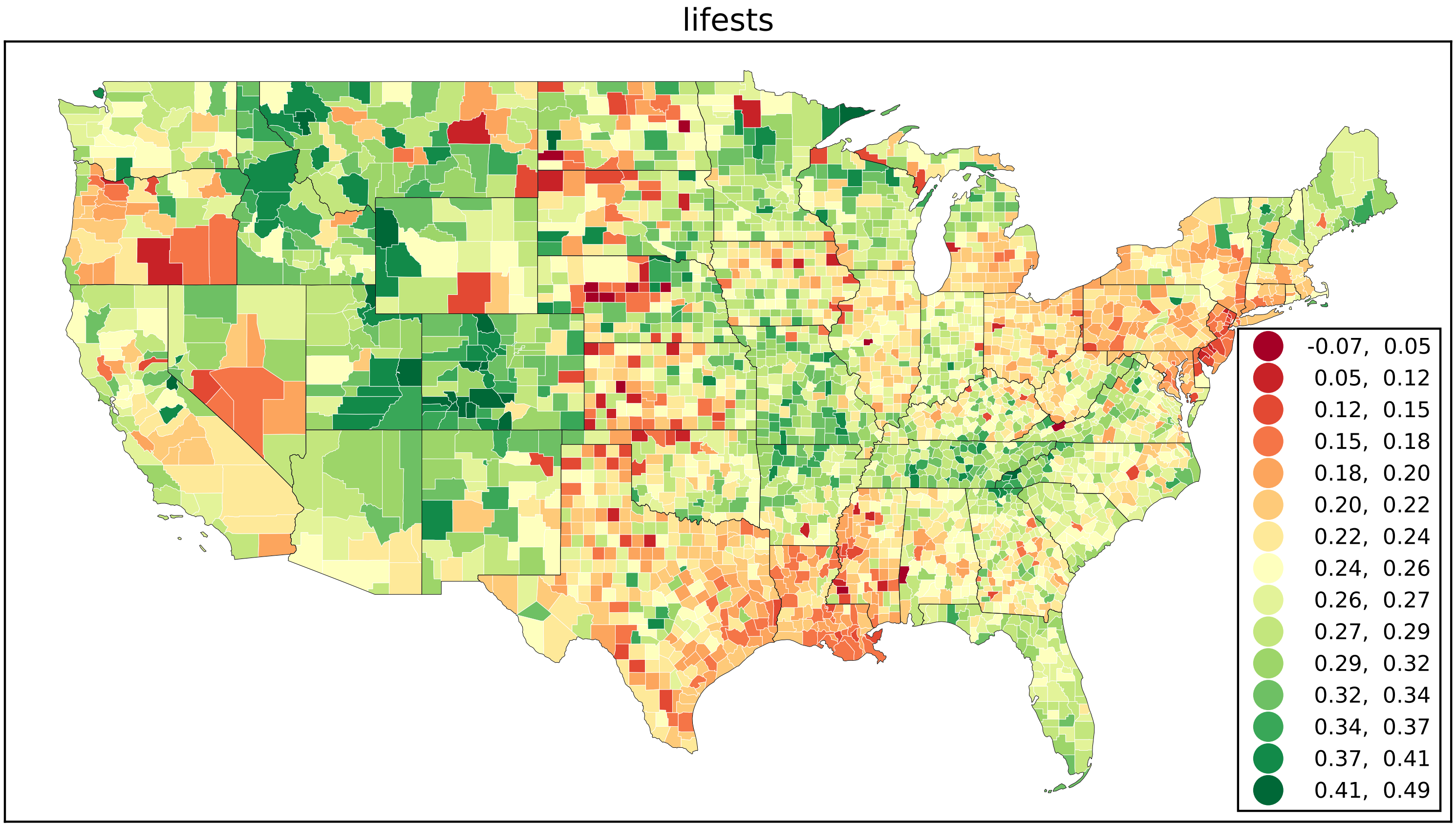

Figure 7: more classes This figure was pulled directly from a web URL.

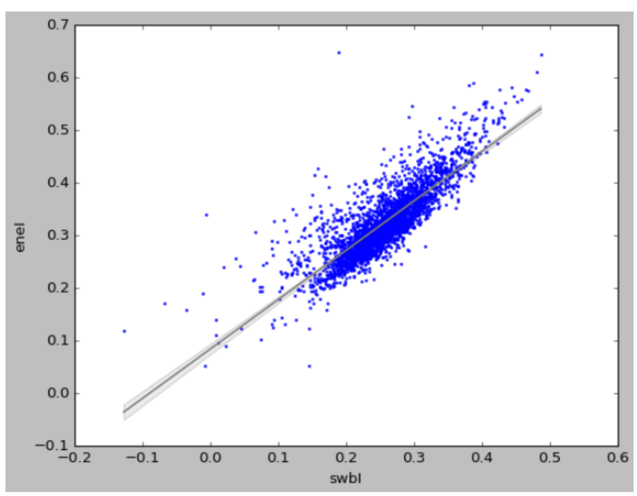

Figure 8: Correlation is 0.83.

STNAME	CTYNAME	POPESTIMATE2015	enel	posfunct	vitality	energy	swbl	life
Montana	Golden Valley County	818	0.650000	0.640000	0.700000	0.600000	0.190000	-0.020
Colorado	Jackson County	1352	0.640000	0.610000	0.680000	0.640000	0.490000	0.430
Colorado	Hinsdale County	762	0.610000	0.620000	0.650000	0.560000	0.480000	0.400
Montana	Musselshell County	4631	0.590000	0.590000	0.570000	0.610000	0.390000	0.240
Wyoming	Sublette County	10064	0.590000	0.570000	0.610000	0.580000	0.380000	0.190
Alaska	Skagway Municipality	1090	0.580000	0.570000	0.620000	0.550000	0.450000	0.400
Wyoming	Teton County	23083	0.580000	0.550000	0.600000	0.590000	0.470000	0.380
California	Alpine County	1080	0.580000	0.570000	0.610000	0.550000	0.430000	0.340
Utah	Rich County	2301	0.580000	0.570000	0.590000	0.560000	0.470000	0.420
Colorado	Saguache County	6250	0.560000	0.570000	0.530000	0.590000	0.450000	0.350
Colorado	Ouray County	4611	0.560000	0.530000	0.560000	0.590000	0.460000	0.380
Colorado	Dolores County	1980	0.550000	0.590000	0.570000	0.500000	0.410000	0.380
Colorado	San Juan County	690	0.550000	0.570000	0.570000	0.530000	0.450000	0.410
Utah	Garfield County	4966	0.550000	0.560000	0.570000	0.530000	0.400000	0.280
Minnesota	Cook County	5249	0.550000	0.510000	0.570000	0.570000	0.440000	0.360
Montana	Madison County	8033	0.550000	0.550000	0.540000	0.570000	0.400000	0.350
Washington	San Juan County	16155	0.550000	0.530000	0.540000	0.580000	0.410000	0.300
Montana	Treasure County	680	0.550000	0.570000	0.530000	0.540000	0.300000	0.240
Tennessee	Hancock County	6577	0.540000	0.510000	0.530000	0.590000	0.360000	0.240
Colorado	Mineral County	742	0.540000	0.550000	0.580000	0.500000	0.420000	0.380
Colorado	Grand County	14742	0.540000	0.510000	0.560000	0.560000	0.410000	0.280
Tennessee	Perry County	7855	0.540000	0.610000	0.520000	0.490000	0.400000	0.350
Hawaii	Kauai County	71052	0.540000	0.510000	0.560000	0.540000	0.430000	0.320
Hawaii	Maui County	163993	0.530000	0.510000	0.550000	0.540000	0.430000	0.310
Utah	Daggett County	1107	0.530000	0.500000	0.590000	0.510000	0.380000	0.340
Wyoming	Lincoln County	18788	0.530000	0.520000	0.520000	0.550000	0.370000	0.250
Idaho	Boise County	6963	0.530000	0.510000	0.530000	0.530000	0.390000	0.270
Colorado	Custer County	4434	0.530000	0.530000	0.490000	0.560000	0.430000	0.340
Montana	Powder River County	1768	0.520000	0.530000	0.550000	0.490000	0.290000	0.160
Colorado	San Miguel County	7850	0.520000	0.510000	0.520000	0.540000	0.430000	0.350
Utah	Kane County	7045	0.520000	0.510000	0.530000	0.530000	0.410000	0.320
California	Mariposa County	17653	0.520000	0.510000	0.540000	0.510000	0.390000	0.350
Utah	Grand County	9530	0.520000	0.520000	0.530000	0.510000	0.430000	0.350
Montana	Stillwater County	9419	0.520000	0.510000	0.470000	0.580000	0.340000	0.210
Minnesota	Clearwater County	8794	0.520000	0.530000	0.460000	0.570000	0.390000	0.280
Wisconsin	Price County	13577	0.520000	0.500000	0.500000	0.560000	0.380000	0.250
Colorado	Costilla County	3576	0.520000	0.500000	0.490000	0.570000	0.430000	0.340
Alaska	Haines Borough	2519	0.520000	0.500000	0.570000	0.480000	0.360000	0.330
Minnesota	Lake County	10540	0.520000	0.510000	0.510000	0.540000	0.400000	0.370
Colorado	Lake County	7455	0.520000	0.490000	0.530000	0.540000	0.420000	0.290
Colorado	Pitkin County	17962	0.520000	0.490000	0.490000	0.570000	0.420000	0.320
Montana	Petroleum County	475	0.520000	0.560000	0.450000	0.530000	0.370000	0.190
Colorado	Crowley County	5633	0.510000	0.500000	0.460000	0.580000	0.360000	0.300
North Dakota	Billings County	950	0.510000	0.500000	0.500000	0.540000	0.420000	0.340
North Carolina	Swain County	14325	0.510000	0.510000	0.520000	0.500000	0.420000	0.290
New Mexico	Catron County	3480	0.510000	0.480000	0.500000	0.550000	0.400000	0.400
Nebraska	Boyd County	2004	0.510000	0.550000	0.460000	0.520000	0.430000	0.350
Idaho	Adams County	3891	0.510000	0.490000	0.510000	0.520000	0.370000	0.250
Colorado	Baca County	3554	0.510000	0.480000	0.540000	0.500000	0.350000	0.280
Nebraska	Garfield County	2013	0.500000	0.490000	0.500000	0.530000	0.380000	0.270
Virginia	Rappahannock County	7409	0.500000	0.510000	0.440000	0.560000	0.400000	0.270
New York	Hamilton County	4700	0.500000	0.450000	0.500000	0.560000	0.360000	0.180
Hawaii	Hawaii County	196111	0.500000	0.490000	0.520000	0.500000	0.370000	0.300
Montana	Flathead County	95940	0.500000	0.500000	0.450000	0.550000	0.410000	0.270
Montana	Meagher County	1822	0.500000	0.520000	0.460000	0.510000	0.310000	0.220
Nevada	Storey County	3881	0.500000	0.480000	0.430000	0.580000	0.400000	0.270
Washington	Skamania County	11386	0.500000	0.500000	0.480000	0.500000	0.380000	0.290
South Dakota	Jackson County	3281	0.490000	0.500000	0.480000	0.500000	0.370000	0.300
Colorado	Summit County	30064	0.490000	0.460000	0.490000	0.530000	0.390000	0.270
South Dakota	Custer County	8456	0.490000	0.470000	0.480000	0.530000	0.390000	0.350
Idaho	Idaho County	16312	0.490000	0.500000	0.450000	0.530000	0.390000	0.290
California	Sierra County	2990	0.490000	0.510000	0.490000	0.480000	0.320000	0.320
Colorado	Clear Creek County	9246	0.490000	0.510000	0.550000	0.410000	0.370000	0.350
Oregon	Wallowa County	6810	0.490000	0.560000	0.550000	0.360000	0.370000	0.300
Utah	Wayne County	2689	0.490000	0.490000	0.480000	0.490000	0.410000	0.350
Vermont	Lamoille County	25265	0.490000	0.480000	0.460000	0.530000	0.400000	0.260
Nevada	Pershing County	6620	0.490000	0.480000	0.430000	0.560000	0.310000	0.140
Montana	Carbon County	10404	0.490000	0.460000	0.480000	0.520000	0.350000	0.230

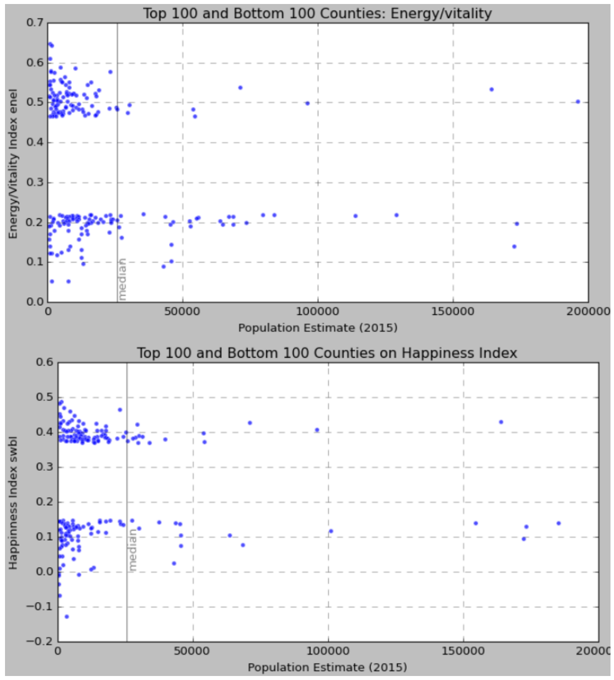

Figure 9: Highest and lowest bot energy/vitality and happiness tends to be in small counties, below median population size.

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